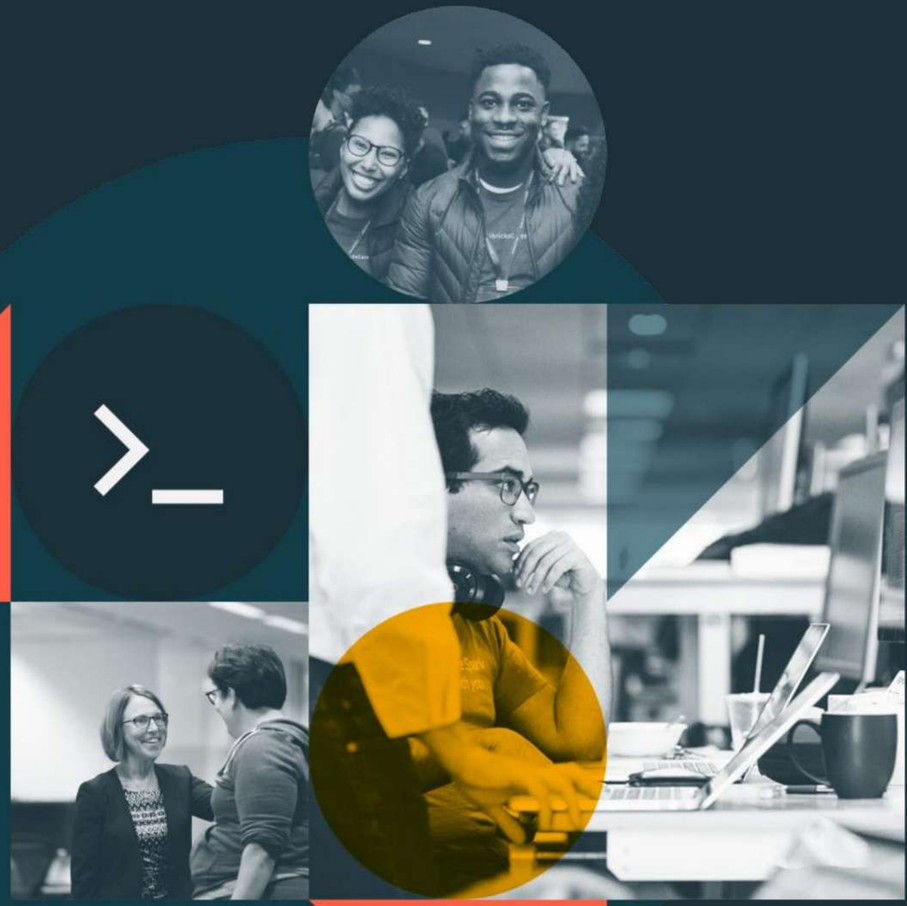
Deploy Workloads with Databricks Workflows



Module 05

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Module Agenda

Deploy Workloads with Databricks Workflows

Introduction to Workflows

Building and Monitoring Workflow Jobs

DE 5.1 - Scheduling Tasks with the Jobs UI

DE 5.2L - Jobs Lab

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Introduction to Workflows

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Course Objectives

1. Describe the main features and use cases of Databricks Workflows
2. Create a task orchestration workflow composed of various task types
3. Utilize monitoring and debugging features of Databricks Workflows
4. Describe workflow best practices

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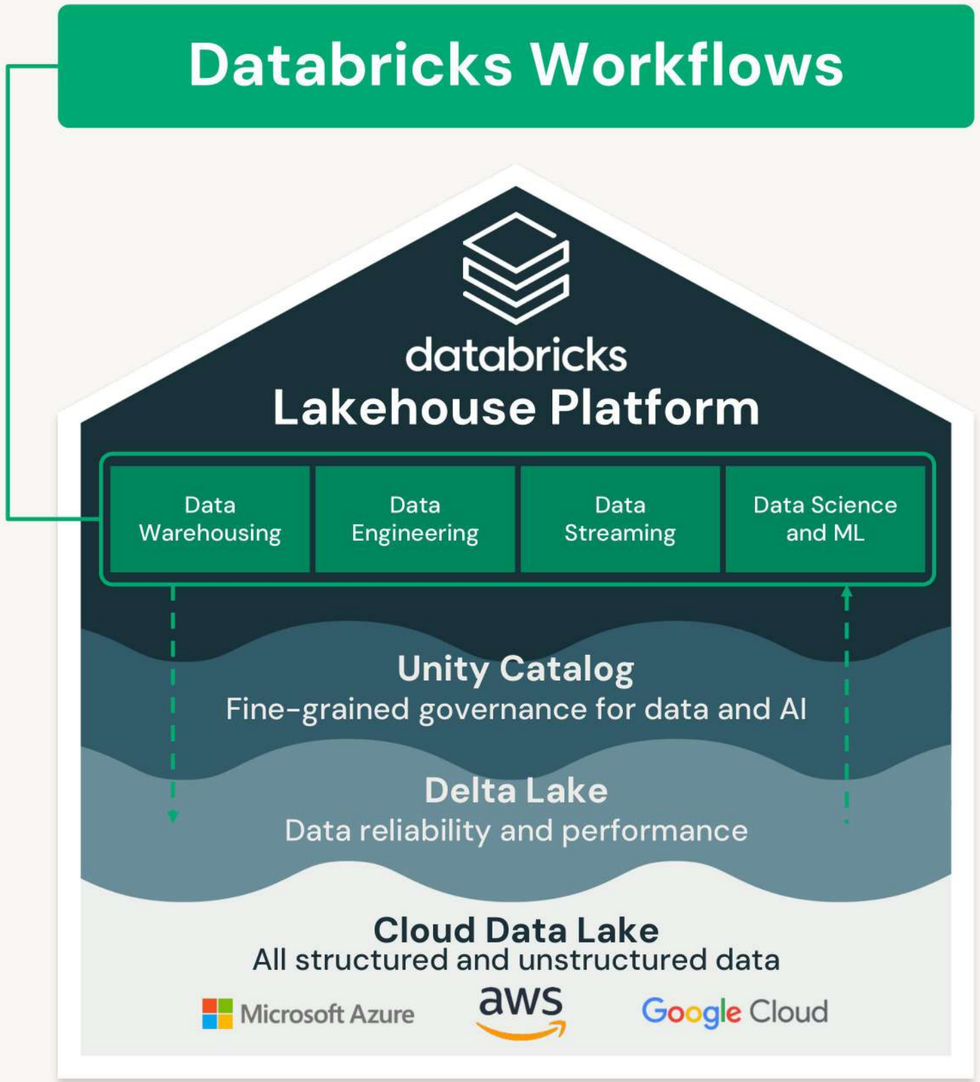
Databricks Workflows

Workflows is a **fully-managed cloud-based general-purpose task orchestration service** for the entire Lakehouse.

Workflows is a service for data engineers, data scientists and analysts to build reliable data, analytics and AI workflows on any cloud.



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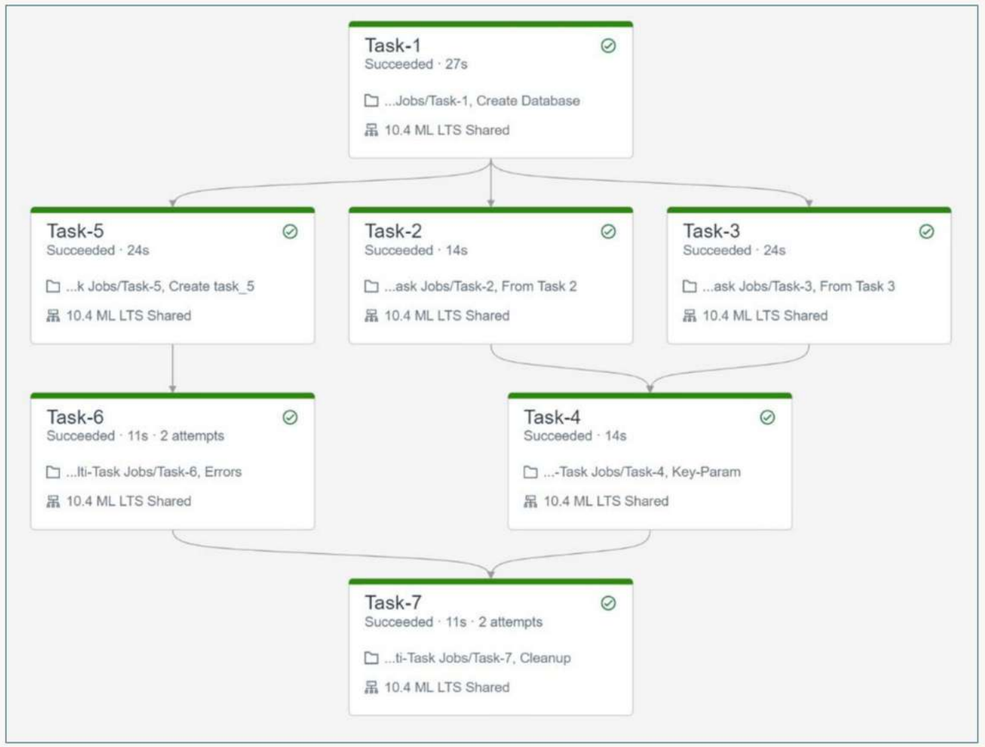
Databricks Workflows

Databricks has two main task orchestration services:

- **Workflow Jobs (Workflows)**
  - Workflows for every job
- **Delta Live Tables (DLT)**
  - Automated data pipelines for Delta Lake



Note: DLT pipeline can be a task in a workflow

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DLT versus Workflow Jobs

Considerations

	Delta Live Tables	Workflow Jobs
Source	Notebooks only	JARs, notebooks, DLT, application written in Scala, Java, Python
Dependencies	Automatically determined	Manually set
Cluster	Self-provisioned	Self-provisioned or existing
Timeouts and Retries	Timeouts not supported Retries handled automatically (in production mode)	Supported
Import Libraries	Not supported	Supported

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DLT versus Jobs

Use Cases

Orchestration of Dependent Jobs

Jobs running on schedule, containing dependent tasks/steps

Jobs Workflows

Machine Learning Tasks

Run MLflow notebook task in a job

Jobs Workflows

Arbitrary Code, External API Calls, Custom Tasks

Run tasks in a job which can contain Jar file, Spark Submit, Python Script, SQL task, dbt

Jobs Workflows

Data Ingestion and Transformation

ETL jobs, Support for batch and streaming, Built in data quality constraints, monitoring & logging

Delta Live Tables

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Workflows Features

Part 1 of 2

Orchestrate Anything Anywhere

Run diverse workloads for the full data and AI lifecycle, on any cloud. Orchestrate;

- Notebooks
- Delta Live Tables
- Jobs for SQL
- ML models, and more

Fully Managed

Remove operational overhead with a fully managed orchestration service enabling you to focus on your workflows not on managing your infrastructure

Simple Workflow Authoring

An easy point-and-click authoring experience for all your data teams not just those with specialized skills

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Workflows Features

Part 2 of 2

Deep Platform Integration

Designed and built into your lakehouse platform giving you deep monitoring capabilities and centralized observability across all your workflows

Proven Reliability

Have full confidence in your workflows leveraging our proven experience running tens of millions of production workloads daily across AWS, Azure, and GCP

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How to Leverage Workflows

- Allows you to build simple ETL/ML task orchestration
- Reduces infrastructure overhead
- Easily integrate with external tools
- Enables non-engineers to build their own workflows using simple UI
- Cloud-provider independent
- Enables re-using clusters to reduce cost and startup time

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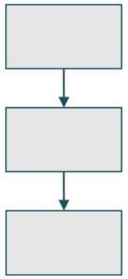
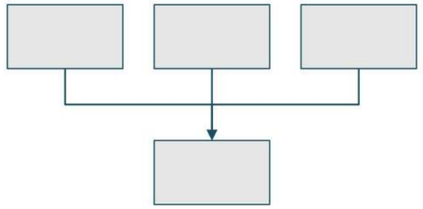
Common Workflow Patterns

Sequence



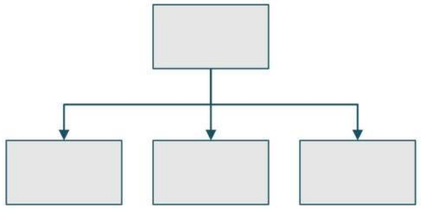
Sequence
- Data transformation/ processing/cleaning
- Bronze/silver/gold tables

Funnel



Funnel
- Multiple data sources
- Data collection

Fan-out



Fan-out, star pattern
- Single data source
- Data ingestion and distribution

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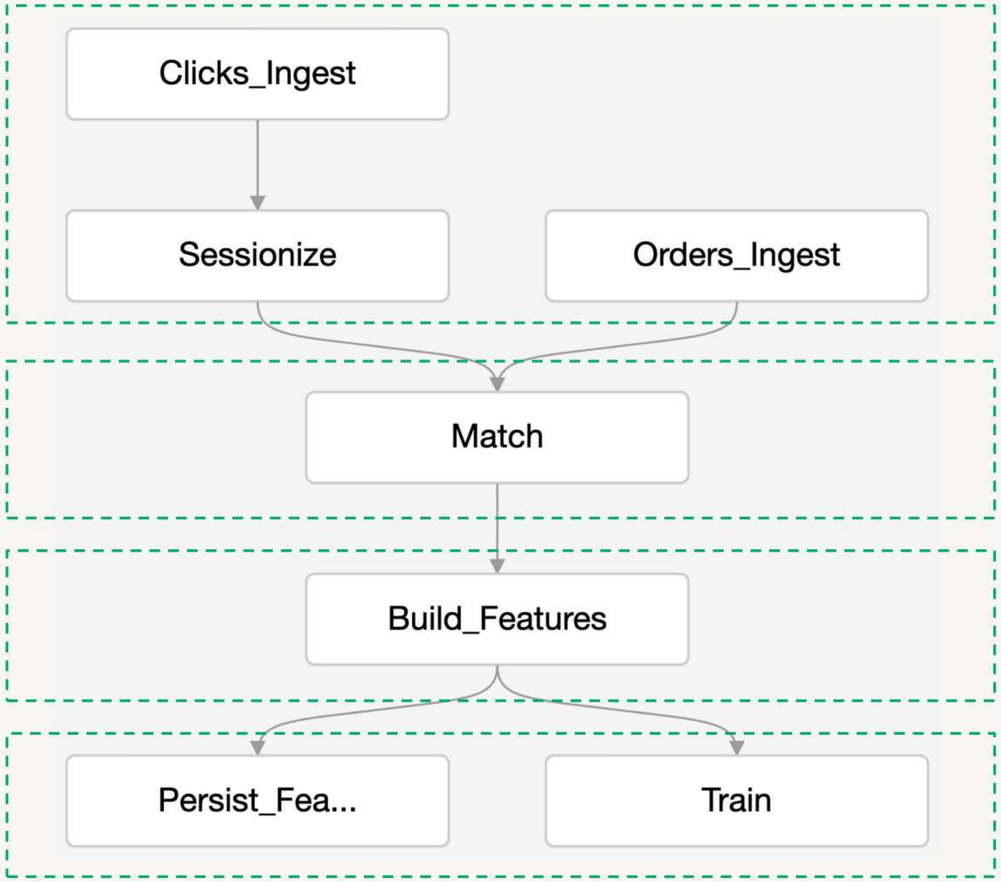
Example Workflow



Data ingestion funnel
E.g. Auto Loader, DLT

Data filtering, quality assurance, transformation
E.g. DLT, SQL, Python

ML feature extraction
E.g. MLflow

Persisting features and training prediction model

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Building and Monitoring Workflow Jobs

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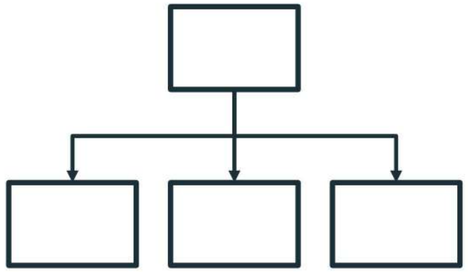
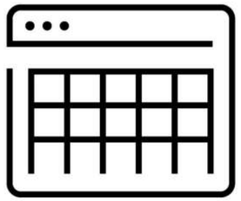
Workflow Job Components

TASKS



What?

SCHEDULE



When?

CLUSTER

How?

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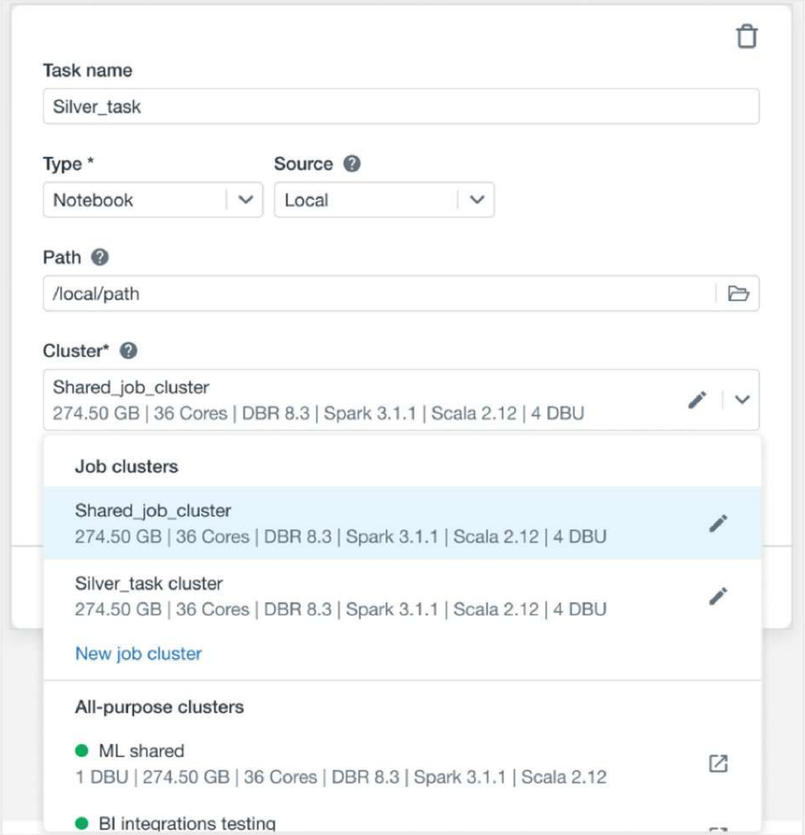
Creating a Workflow

Task Definition

While creating a task;

- Define the task type
- Choose the cluster type
  - Job clusters and All-purpose clusters can be used.
  - A cluster can be used by multiple tasks. This reduces cost and startup time.
- If you want to create a new cluster, you must have required permissions.
- Define task dependency if task depends on another task



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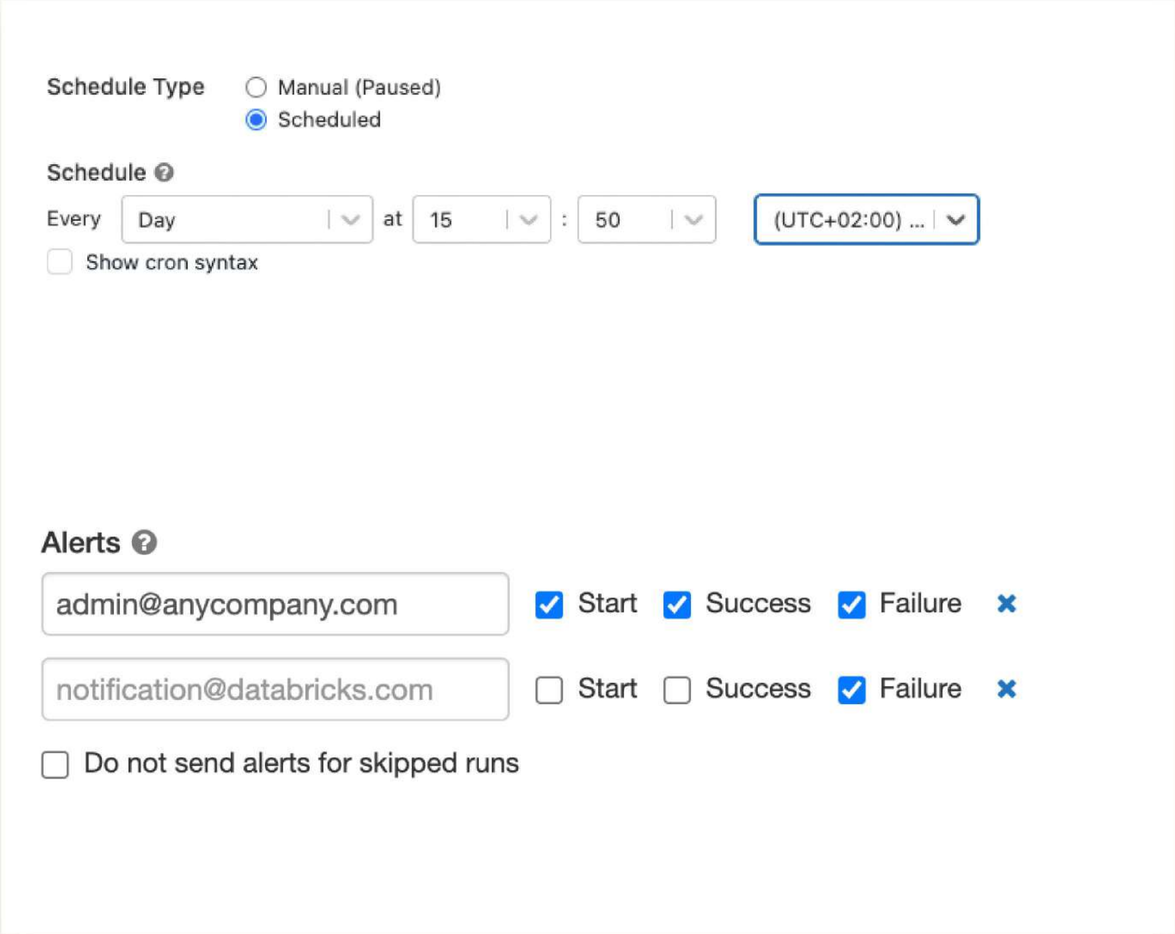
Monitoring and Debugging

Scheduling and Alerts

You can run your jobs **immediately** or **periodically** through an easy-to-use scheduling system.

You can specific alerts to be notified when runs of a job **begin, complete or fail**. Notifications can be sent via email, Slack or AWS SNS.



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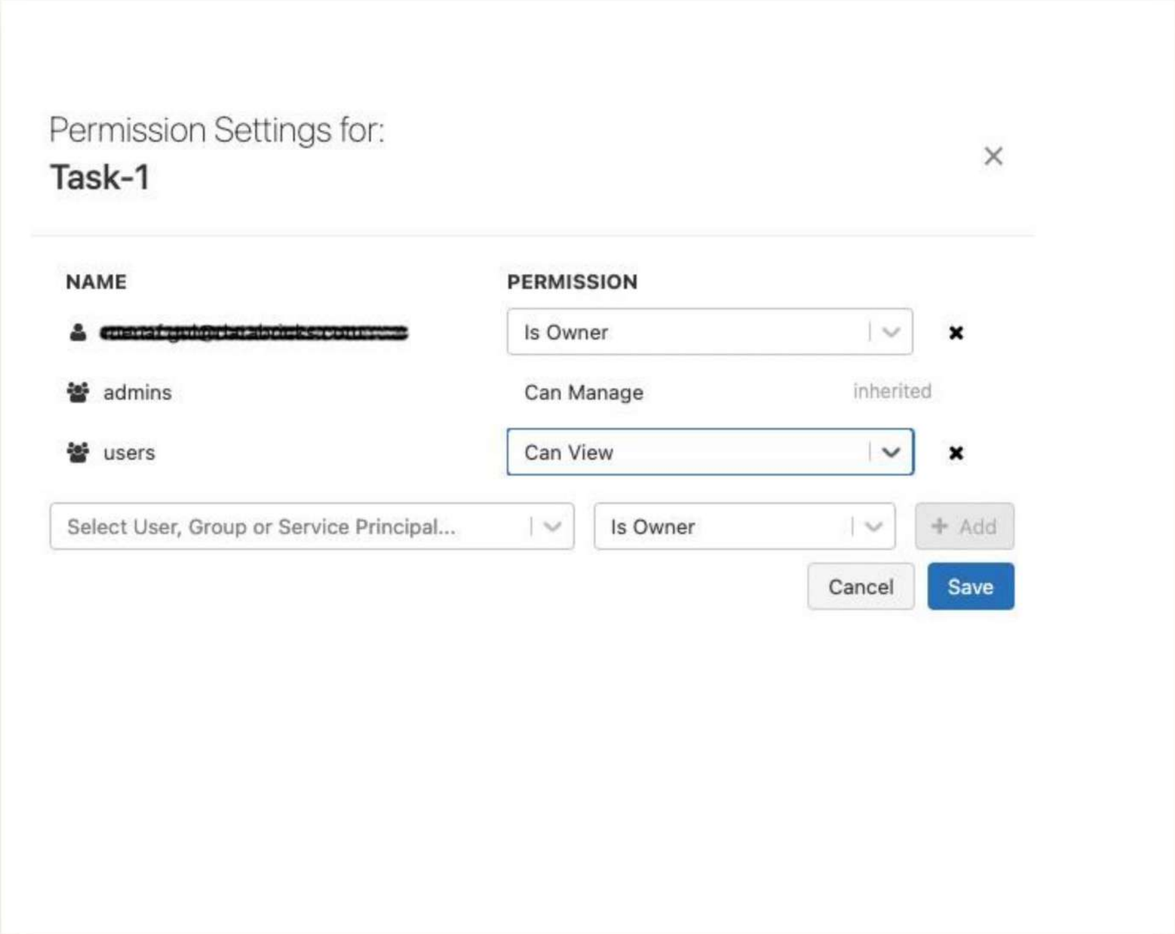
Monitoring and Debugging

Access Control

Workflows integrates with existing resources access controls, enabling you to easily manage access across different teams.



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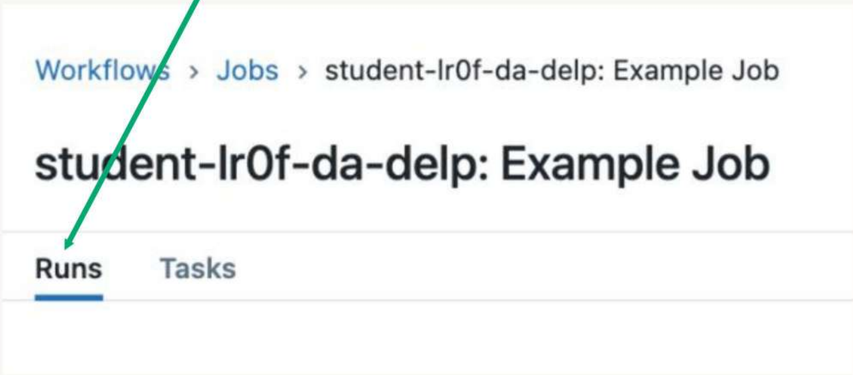
Monitoring and Debugging

Job Run History

Workflows keeps track of job runs and save information about the success or failure of each task in the job run.

Navigate to the **Runs** tab to view completed or active runs for a job.



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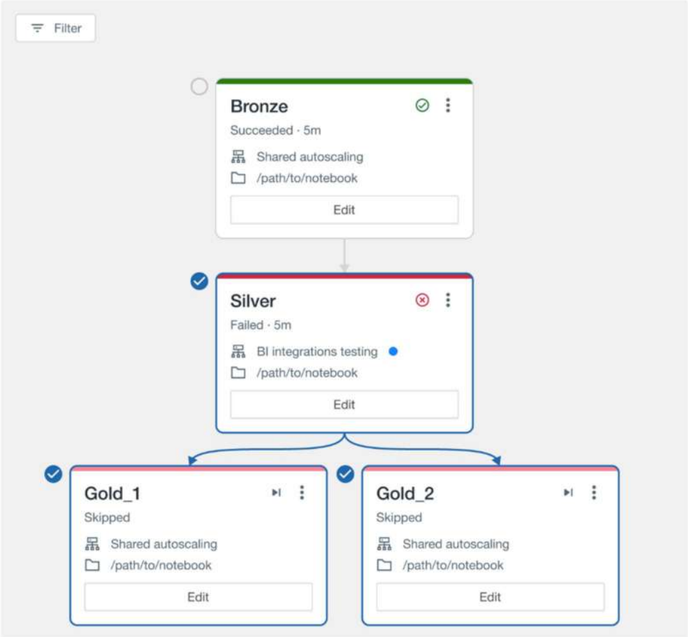
Monitoring and Debugging

Repair a Failed Job Run

Repair feature allows you to re-run only the failed task and sub-tasks, which reduces the time and resources required to recover from unsuccessful job runs.



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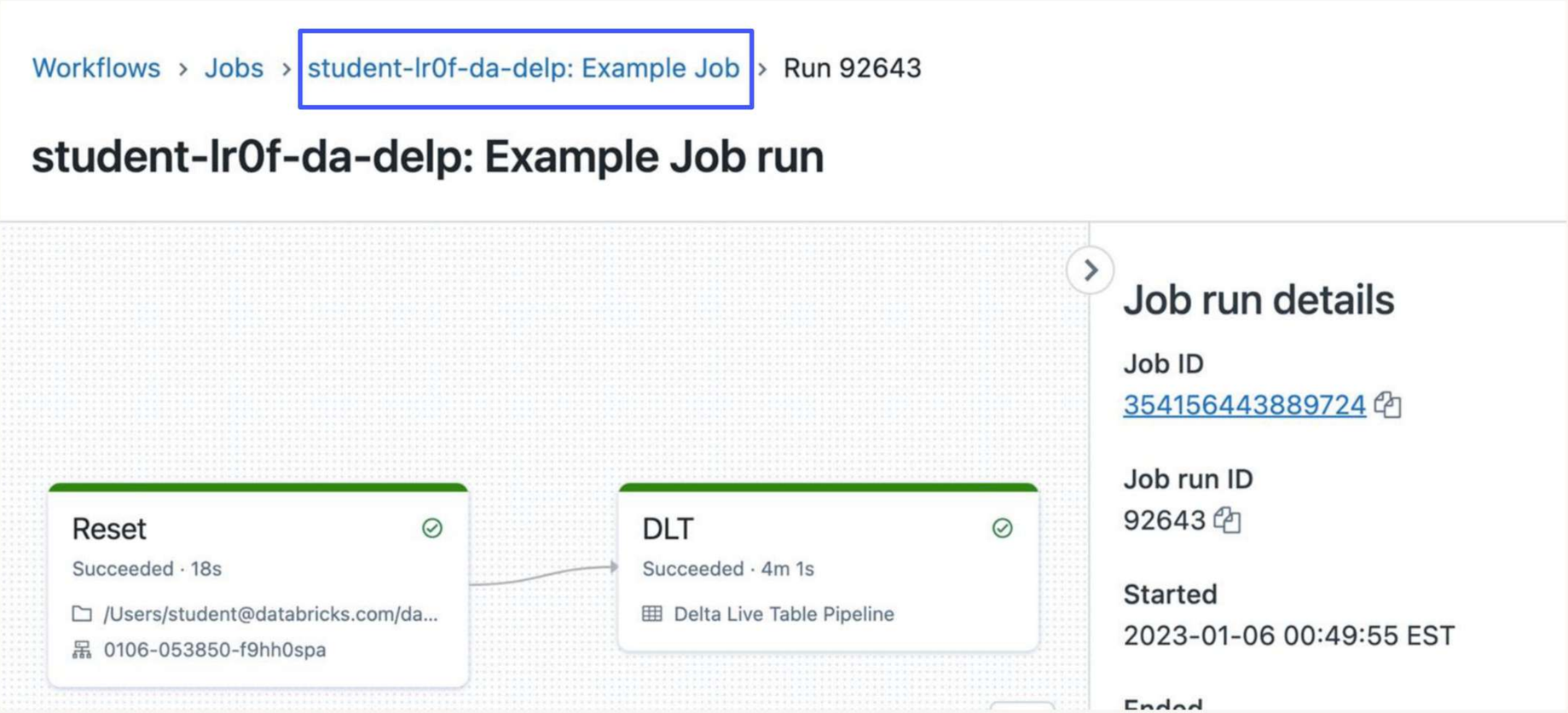
Navigating the Jobs UI

Use breadcrumbs to navigate back to your job from a specific run page



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Navigating the Jobs UI

Runs vs Tasks tabs on the job page

Use **Runs** tab to view completed or active runs for the job

Use **Tasks** tab to modify or add tasks to the job

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DE 5.1.1: Task Orchestration

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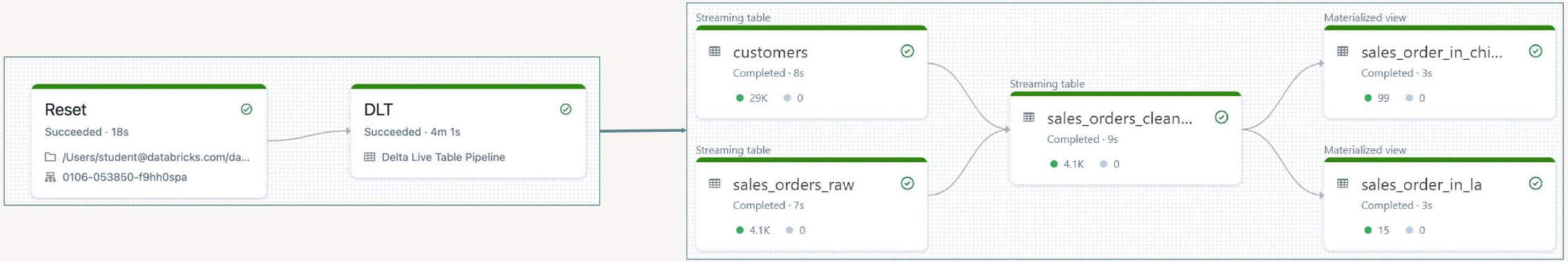
Demo: Task Orchestration

DE 5.1.1 - Task Orchestration

- Schedule a notebook task in a Databricks Workflow Job
- Describe job scheduling options and differences between cluster types
- Review Job Runs to track progress and see results
- Schedule a DLT pipeline task in a Databricks Workflow Job
- Configure dependency between tasks via Databricks Workflows UI



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DE 5.2.1.L: Task Orchestration Lab

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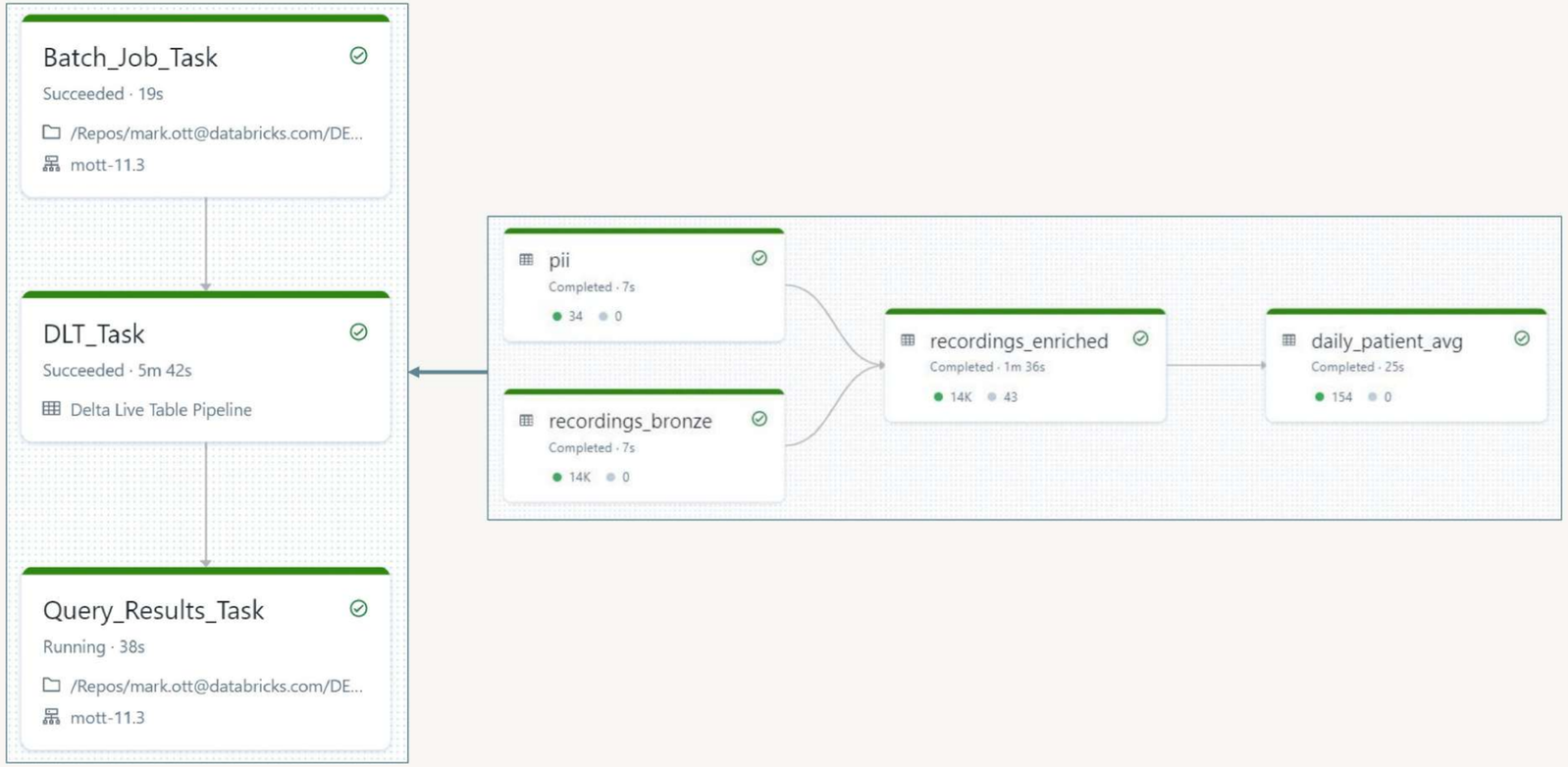
Lab: Task Orchestration

DE 5.2.1.L - Task Orchestration



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